

Smart Beehive Monitoring System using IoT

Vijay Gaikwad
Department of Multidisciplinary Engineering
Vishwakarma Institute of Technology
Pune, India
vijay.gaikwad@vit.edu

Amruta Amune
Department of Information Technology
Vishwakarma Institute of Technology
Pune, India
amruta.amune@vit.edu

Vaishali Rajput
Department of Computer Engineering
Vishwakarma Institute of Technology
Pune, India
vaishali.rajput@vit.edu

Vinayak Musale
Department of Computer Engineering and
Technology
Dr. Vishwanath Karad MIT World Peace
University, Pune, India
vinayak.musale@gmail.com

Neha Rajas
Department of AI & DS
Vishwakarma Institute of Technology
Pune, India
neha.rajas@vit.edu

Surabhi Kakade
Department of AI & DS
Vishwakarma Institute of Technology
Pune, India
surabhi.kakade@vit.edu

Abstract— Bees are the main source of pollinators for various wild plant species and is the only source of honey production. Beehive management frequently faces challenges such fluctuating weather conditions, ambiguous indicators of hive health, and the requirement for timely honey extraction. The proposed initiative uses Internet of Things (IoT) technologies to address the urgent issues that modern beekeepers are facing. Our technology uses sensors to track temperature, humidity, hive weight, and bee count inside beehives in order to overcome these challenges. Beekeepers can take proactive steps to preserve ideal circumstances by monitoring environmental data on a regular basis. The Beehive Monitoring System provides beekeepers remote accessibility. This Paper presents a thorough analysis of the system's methodology and results, emphasizing its potential to transform beekeeping methods. The proposed system also helps us find the most appropriate conditions for optimal honey production. This includes a temperature of 30 to 36 degree Celsius and a relative humidity of 40% to 60% for the prototype tested.

Keywords— Apiculture, IoT, Monitoring, Remote Sensing, Smart System

I. INTRODUCTION

Bee farming, also known as apiculture, involves the cultivation and management of honeybees for various purposes, primarily for the production of honey, beeswax, and other bee-related products. It's an age-old practice that has been adopted worldwide for its numerous benefits. Bee farming offers multiple benefits for farmers worldwide, especially in India. Through pollination, bees boost crop yields, benefiting fruits, vegetables, and cash crops like cotton. Additionally, honey production provides a valuable source of income, meeting both domestic and international demand. Integrating bee farming diversifies revenue streams, reducing dependency on single crops and mitigating market risks. Furthermore, maintaining bee colonies contributes to biodiversity conservation and ecological balance by aiding in the reproduction of wild plants. Overall, bee farming enhances productivity, profitability, and environmental sustainability for farmers, making it a crucial component of modern agricultural practices.

Bee farming faces significant challenges in remote monitoring and management. Limited monitoring capabilities hinder early detection of issues like diseases and pests, compromising colony health. Without effective tools, farmers struggle to manage colonies efficiently, leading to productivity losses and increased risks. Moreover, the lack of real-time data impedes decision-making, making it hard to implement timely interventions and optimize operations. These challenges underscore the urgent need for advanced monitoring systems to enhance bee farming practices, ensuring the sustainability and profitability of this vital agricultural sector. In addition to the challenges mentioned, bee farming also grapples with environmental factors such as climate change and habitat loss, which can adversely affect bee health and foraging patterns. Climate variability can disrupt flowering seasons and alter the availability of nectar and pollen, impacting honey production and colony development. Furthermore, habitat degradation and pesticide use pose significant threats to bee populations, leading to declines in pollination services and biodiversity loss.

The reduction in bee numbers worldwide is a noteworthy obstacle for agricultural practices, food security, and ecosystems. Beekeeping, which is essential for the preservation of pollinators, is fraught with difficulties, such as the requirement for efficient hive management and unpredictability in the environment. We have developed a project called the "Beehive Monitoring System," an Internet of Things-based solution, to address these issues and improve beekeepers' capacity to track hive conditions in real-time. The goal of our effort is to offer a useful strategy for hive management and conservation in light of worries about declining bee numbers. Beekeepers can receive vital insights for prompt intervention and optimal hive health with the Beehive Monitoring System, which uses Internet of Things (IoT) to provide continuous monitoring of temperature, humidity, hive weight, and bee count.

In remote areas lacking reliable internet connectivity, implementing alternative connectivity solutions such as satellite or low-power wide-area network (LPWAN)

technologies ensures continuous data transmission from beehives to the central monitoring system. While LPWAN technologies like LoRaWAN or NB-IoT offer long-range connectivity with minimal power consumption, perfect for distant regions, satellite communication enables worldwide coverage. In locations where regular internet connectivity is not readily available, these solutions allow for the real-time monitoring and administration of beehives.

The objectives that we serve as the guiding principles are as follows:

- To develop an monitoring system equipped with technology aimed at providing real-time insights into bee colony health, thereby enabling early detection and proactive management of potential threats.
- To design and implement a user-friendly mobile application that empowers beekeepers with intuitive tools for remote monitoring.
- Foster collaboration and knowledge-sharing within the beekeeping community by creating a platform for information exchange, best practices dissemination, and peer support.

The overview of this paper is as follows. The Section I consists of introduction to the apiculture, the problems faced by existing systems and the need of beehive monitoring system. In the section II we discuss about the methodology used for the system. Section III comprises of the detailed description of each research paper and the problems, solutions, insights, and results from each of the shortlisted papers. In section IV, we present the results/outcome of our system. In section V, we have discussed about the future prospects and improvements. Finally, we conclude and summarize our study in the section V.

II. LITERATURE SURVEY

In [1], A. Terenzi and S. Cecchi described about a system that helps in monitoring various parameters related to Beehive. The parameters included here are temperature, sound, CO₂ and humidity to assess the health and productivity of honey bees. Total 4 load cells are used in this paper and are attached by a Wheatstone bridge. The data from the sensors is gathered, processed, and sent to a remote server via a Python script. By examining the weight variations and their relationship to other factors, the system is evaluated in an actual setting and demonstrates the ability to identify various events, including as swarming, honey collection, and normal activity.

In [2, 12, 14], Sherry Chalotra explained Bee Colony Optimization (BCO). BCO is a potent meta-heuristic algorithm that was inspired by honey bee foraging behavior. It is a useful tool for optimization issues because of its capacity to replicate swarm intelligence and maneuver through intricate solution spaces. Scholars have investigated a number of BCO variations, including the Bee Algorithm, which uses robotic bees to explore solutions. Applications are found in many different fields, including as scheduling, categorization, transportation, and optimization. Beyond

individual decision-making, BCO continues to be a source of inspiration for computational intelligence.

In [3], Fiona Edwards Murphy and Michele Magno explained about Utilizing wireless sensor networks, the b+WSN: Smart Beehive Monitoring System keeps an eye on temperature, humidity, gas levels, and bee activity in beehives. Perspectives from the fields of biology, environment, meteorology, and engineering are revealed by the multidisciplinary investigation. Notably, suggested algorithms evaluate colony health, anticipate weather, and automatically determine hive status. The Libelium Waspote platform, a number of sensors, Zigbee, and 3G radios are all used in the system architecture. The document references pertinent research in the domains of wireless sensor networks and beekeeping¹². All things considered, this research uses cutting edge monitoring technologies to better understand and improve beekeeping methods.

In [4], Selcuk ASLAN explained how well an artificial bee colony (ABC) method that has been parallelized performs in addressing the probabilistic sensor deployment issue for wireless sensor networks. Through subcolony division and inter-subcolony communication, the parallel ABC algorithm optimizes the area covered by sensors. According to the results, the parallel approach performs better than the serial version, lowering computation time while obtaining coverage ratios that are comparable to or superior. The paper emphasizes how effective sensor deployment with ABC algorithms may be achieved by utilizing multi-core CPUs and even mobile platforms. In conclusion, the ABC algorithm's capabilities are improved by parallelization, which makes it a viable option for wireless sensor network optimization problems.

In [5], Olivier Debauche explained about keeping an eye on bees using the internet and computers. Bees are in trouble because of something called Colony Collapse Disorder. This happens when many bees die because of things like pesticides, bugs, sickness, and climate changes. This is bad for nature and growing food. Precision beekeeping is like a smart way to take care of bees. It uses special sensors and the internet to watch over bee colonies. These sensors check things like temperature, humidity, weight, and sound. If something is wrong, we can find out early and help the bees stay healthy. The researchers suggest a plan with tools in the bee homes, a cloud computer to save and process data, and a way for scientists to share information. This plan aims to protect bees and make sure they can keep doing their important job of pollinating crops.

In [6], Antonio Rafael Braga explained the need to maintain the health of bees. Bees face numerous challenges, including structural changes in their dwellings, toxic materials, pests, and a shortage of supplies. The study offers a method to determine whether a bee colony is going to experience major problems in order to address this. They employed an intelligent computer system that was trained over a three-year period using actual data from 27 beehives, taking into account variables including weather, weight, and hive temperature. The outcomes demonstrate that this approach can accurately forecast bee colony health with over 90% of the time. This could be a helpful tool for

beekeepers to maintain the health of their hives and make sure the bees remain contributing to plant growth.

In [7,11,13], Rahman Tashakkori explained how beekeepers have always had to physically monitor honey beehives, but this is beginning to change as a result of smart technology like Internet of Things (IoT) devices. This study investigates the tracking of 28 beehives in North Carolina with the help of Internet of Things devices and the AppMAIS project's Thingsboard dashboard. A camera records bee activity, a microphone records sounds, a scale keeps track of hive weight, and sensors assess temperature and humidity within each hive. Beekeepers can more easily monitor the health of their hives on a regular basis because all of this data is delivered to the Thingsboard dashboard. This study demonstrates the effectiveness of employing Internet of Things (IoT) solutions, such as Thingsboard, to monitor honey beehives and make necessary adjustments.

In [8], Asaduz Zaman explained how bees play a crucial role in pollinating crops and creating globally significant products like honey, which is why hive monitoring is becoming more and more popular. The "OIP Framework," a useful framework for classifying sensor-assisted hive monitoring into Operational, Investigative, and Predictive techniques, is proposed in this paper. Every category, from quickly addressing hive health issues to regulating environmental changes, depends on particular technologies and provides distinct advantages. The study highlights the potential benefits for all countries, especially those with lesser incomes, despite the attention that honey bees are currently receiving in wealthier nations. Better monitoring of the hives can help control pollination, respond quickly to pests and illnesses, and even mitigate hazards like fires and floods. The study promotes a thorough and codified approach by highlighting the vital role that bee health plays in maintaining ecosystems and horticulture. It assumes that predictive monitoring will play a major role in the future and will enable beekeepers to use cutting-edge technologies to manage hives proactively.

In [9], author explained although honey bees are essential to our ecosystem and way of life, the winter collapse of bee colonies is a worrying trend. This research focuses on employing a multi-sensor system to monitor the air temperature inside hives in order to assess the health of bee colonies. In contrast to other research, over 30 bee colonies are being continuously monitored for two full winters in this study. The study finds patterns in the variations in air temperature within hives by examining both short-term hourly data and long-term daily average data. The results imply that temperature-derived indicators can be used to evaluate the overwintering health of bee colonies, offering important information for colony management.

In [10], Xinle Duan explained the impact of pyraclostrobin, a fungicide often used in plant care, on the age-related sensitivity of worker honeybees is investigated in this study. A bee colony's social structure and tasks change with age, which affects how the colony reacts to outside stressors like pesticides. The study looks at how pyraclostrobin affects immunological response, food metabolism-related gene expression, and detoxifying and

protecting enzymes in bees of various ages. The findings show that in the presence of pyraclostrobin, older worker bees, particularly foragers at 21 days old, exhibit decreased tolerance, changed enzyme activity, and downregulated gene expression. These results highlight the physiological and biochemical effects of pyraclostrobin on honeybees that are age-dependent and provide important information for determining possible threats to pollinator populations.

III. METHODOLOGY

This section describes the various sensors used and methodology adopted for the implementation of the prototype design for Beehive Monitoring.

1. DHT11

The DHT11 sensor is renowned for its capability to measure both temperature and humidity levels with high accuracy. Operating with a humidity range of 20% to 90% RH (Relative Humidity) and a temperature range of 0°C to 50°C, it provides reliable data crucial for monitoring beekeeping environments. Utilizing a digital signal output, as shown in figure 1, the DHT11 sensor employs a capacitive humidity sensor and a thermistor to measure temperature, ensuring precise readings. This sensor functions by detecting changes in capacitance due to variations in humidity levels and resistance changes in the thermistor in response to temperature fluctuations.

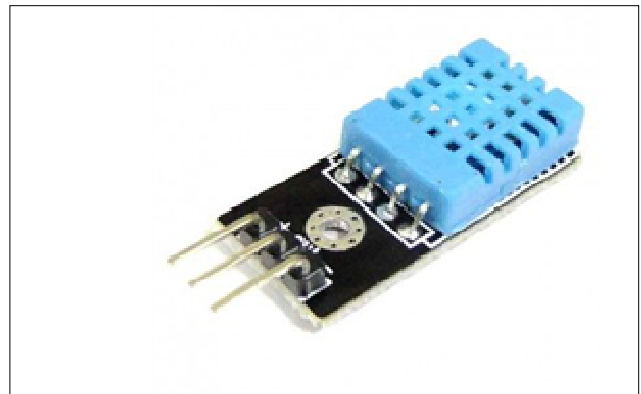


Fig. 1 The DHT11 Sensor for Temperature and Humidity Sensing

2. TTP233P

The TTP233P tactile sensor serves as a vital component for counting bees within the hive. Featuring a low power consumption design and high sensitivity, this sensor detects tactile input, making it ideal for monitoring bee activity. With a testing range suitable for detecting bee movements and hive interactions, the TTP233P ensures accurate bee counting data essential for hive management. Operating on a simple principle, as shown in figure 2, the TTP233P sensor triggers a digital output signal upon tactile contact, enabling precise counting of bee entries and exits from the hive.

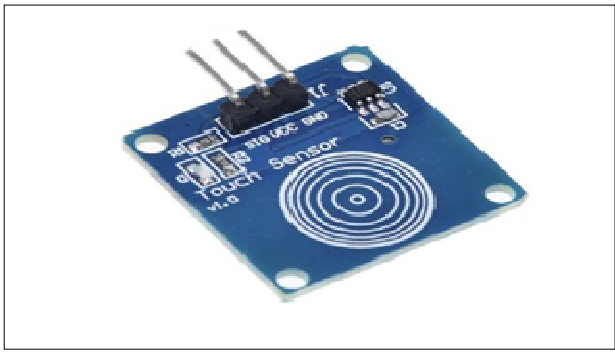


Fig. 2 The TTP223P Sensor for Touch Detection

3. Load Cell

The load cell chosen for hive weight readings offers precision and reliability in measuring the weight of beehives. With a testing range capable of accurately capturing fluctuations in hive weight, this sensor facilitates monitoring hive health and productivity as shown in figure 3. Equipped with strain gauges and employing Wheatstone bridge principles, the load cell converts mechanical force exerted on the hive into electrical signals, allowing for precise weight measurements. By detecting changes in electrical resistance due to applied force, the load cell provides real-time data on hive weight, aiding beekeepers in assessing hive conditions and honey production.

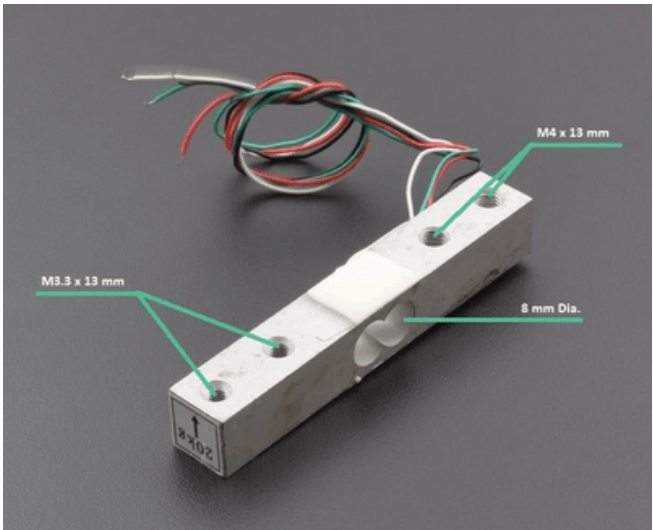


Fig. 3 Load Cell Sensor used for sensing the beehive weight

Sensor calibration and maintenance are critical for optimal performance of the Smart Beehive Monitoring System. Regular calibration ensures accuracy, while maintenance checks ensure sensors are clean and functional. Scheduled calibration intervals and documented procedures help maintain data accuracy and system reliability, enhancing overall performance and effectiveness.

A. Building blocks of the Beehive Monitoring System

The development of the Beehive Monitoring System (BMS) involved the integration of three main components: the IoT

system, the cloud infrastructure utilizing Blynk, and the application created using Unity. Each component played a crucial role in enabling real-time monitoring and management of hive conditions.

1. IoT System

The IoT system comprised hardware components including sensors such as the DHT11 temperature and humidity sensor, TTP223P tactile sensor, and load cell, the NodeMCU ESP8266 microcontroller, and supporting circuitry. The circuit diagram depicted the connections between these components, outlining the flow of data acquisition and transmission. Data from the sensors were collected by the NodeMCU ESP8266, which processed and transmitted the data to the cloud for further analysis and visualization.

2. Cloud Infrastructure using Blynk

The Blynk served as the cloud infrastructure for the BMS, facilitating communication between the IoT devices and the application. Data streams were established within the Blynk platform to receive sensor data from the IoT devices, with customizable data types to accommodate various sensor readings (e.g., temperature, humidity, bee counts, hive weight). However, Blynk had limitations in terms of the number of data streams and the frequency of data updates, necessitating careful consideration of data transmission protocols to optimize efficiency. API calls were utilized to retrieve and visualize data within the Unity application, enabling real-time monitoring of hive conditions.

3. Application using Unity

The Unity application provided a user-friendly interface for beekeepers to access and visualizes hive data. Unity's versatility allowed for the creation of interactive dashboards and visual representations of sensor data, enhancing the user experience. By integrating API calls to the Blynk platform, the application retrieved sensor data in real-time, displaying it to the user in a clear and intuitive manner. This seamless integration between the IoT system, cloud infrastructure, and application ensured that beekeepers could remotely monitor and manage hive conditions with ease.

Several tactics are used in the Smart Beehive Monitoring System to mitigate the environmental impact of employing IoT devices. By using energy-efficient sensors and equipment, carbon emissions related to the use of electricity are reduced by minimizing power consumption. The environmental footprint of devices is further decreased by using eco-friendly materials in their manufacturing and packaging. By putting in place acceptable end-of-life disposal procedures, improper IoT device recycling or disposal is ensured, limiting the buildup of electronic trash. The Smart Beehive Monitoring System promotes sustainable beekeeping practices while minimizing its environmental effect through the integration of

sustainability into its design, production, and disposal processes.

In Smart Beehive Monitoring System, data privacy and security measures are imperative. Sensitive information is shielded from intrusions and breaches by encryption and frequent audits. Open communication through a thorough

privacy policy guarantees that users are aware of how their data is used, which promotes accountability and confidence. Setting data privacy as a top priority protects sensitive data, maintains moral principles and user trust, and reduces the likelihood of data breaches or misuse.

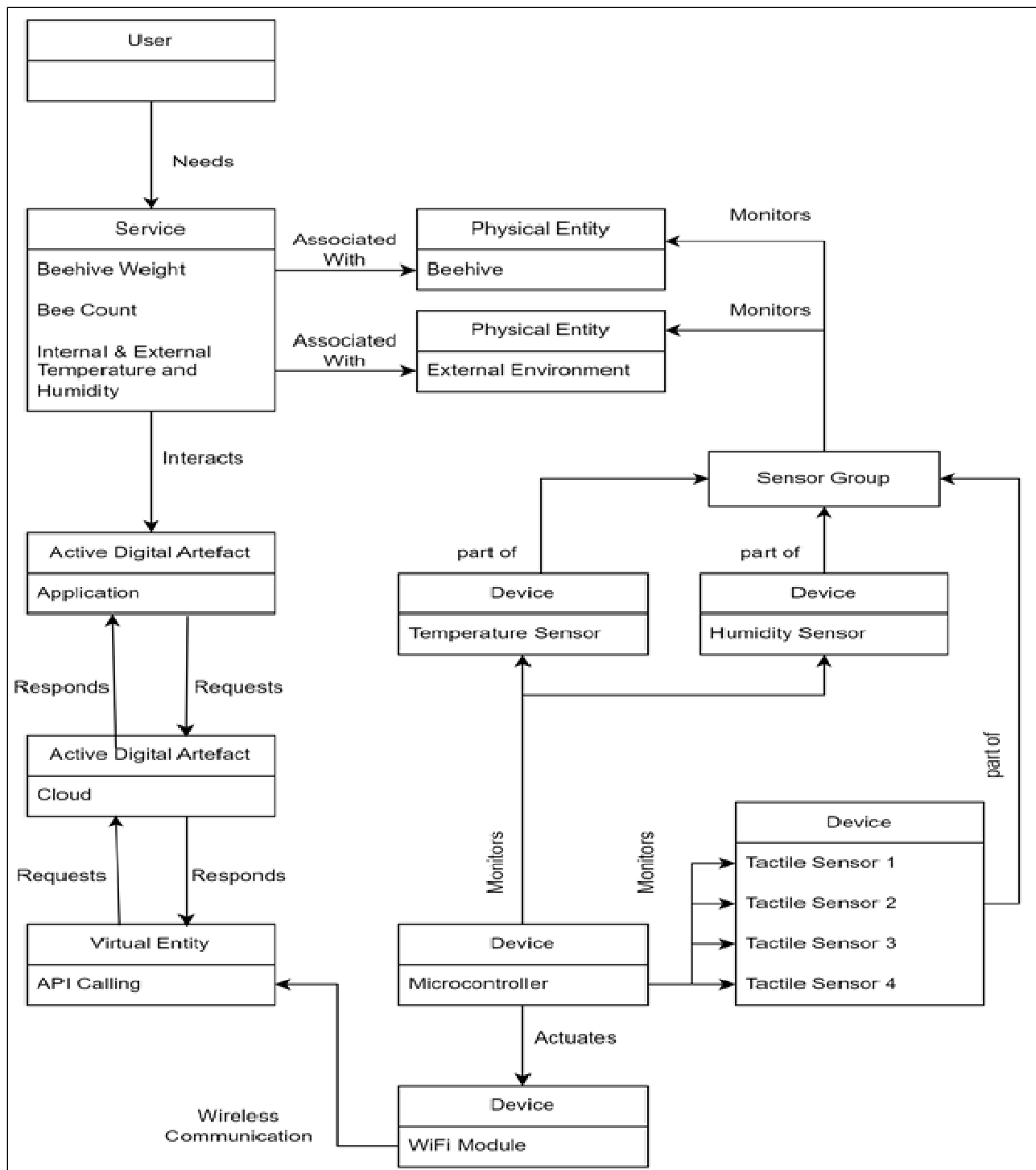


Fig. 4 Process Flow Diagram of Beehive Monitoring System with Entities and their Functions and Properties

The Beehive Monitoring System process flow presented in figure 4 integrates cutting-edge IoT components to address the critical challenges faced by modern beekeepers. This system is designed to provide real-time monitoring and management of beehive conditions, leveraging a network of sensors to collect data on key environmental parameters and bee activities.

The core entities of this system include the DHT11 sensor, which measures temperature and humidity levels within the hive, ensuring optimal conditions for bee health and productivity. The TTP223P tactile sensor accurately counts the number of bees entering and exiting the hive, providing vital data on bee activity. Additionally, the load cell measures hive weight fluctuations, which can indicate changes in honey production and overall hive health. These sensors are interfaced with the NodeMCU ESP8266 microcontroller, which processes and transmits the collected data to the cloud via the Blynk platform. The cloud infrastructure facilitates seamless communication between the IoT devices and the user application, enabling real-time data visualization and remote monitoring. The Unity application offers a user-friendly interface for beekeepers, featuring interactive dashboards that display sensor data and provide actionable insights. This integration of hardware, cloud services, and application software ensures that beekeepers can proactively manage their hives, responding swiftly to any anomalies or adverse conditions.

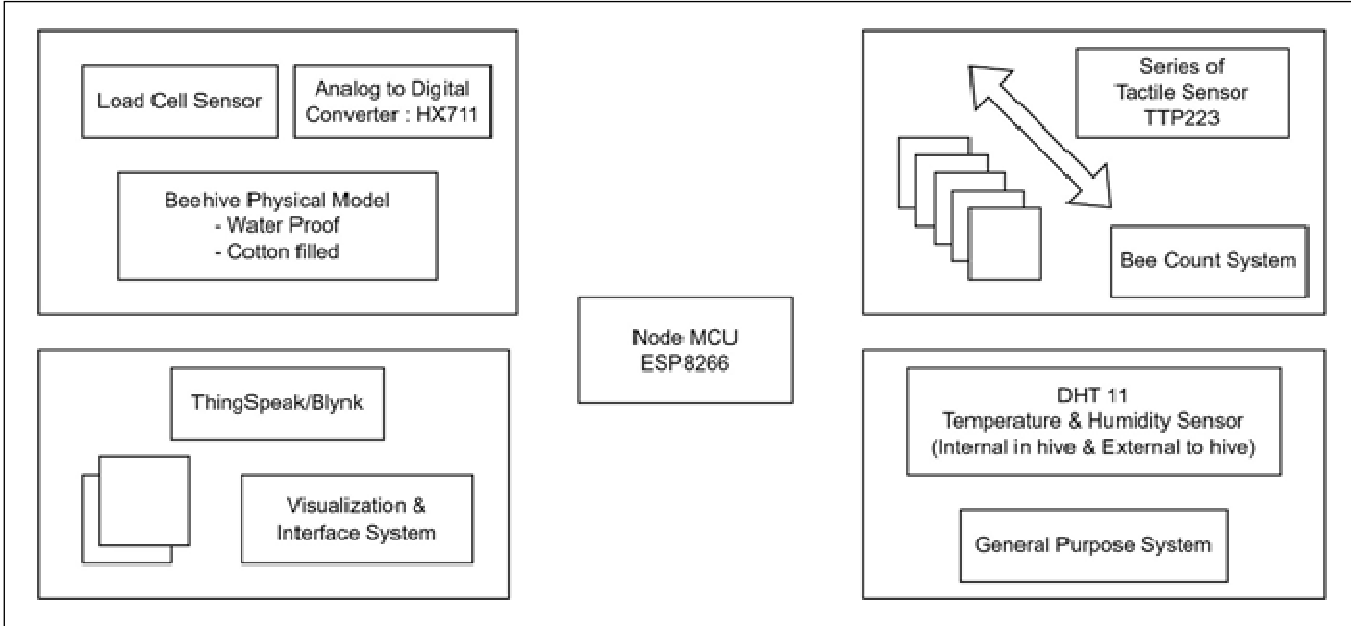


Fig. 5 System Architecture of the Beehive Monitoring System

The System Architecture of the Beehive Monitoring System as shown in figure 5 is meticulously designed to ensure efficient data collection, processing, and real-time monitoring of hive conditions. At the core of the architecture is the NodeMCU ESP8266 microcontroller, which serves as the central hub for data acquisition and transmission. The system utilizes a suite of sensors, including the DHT11 sensor for measuring temperature and humidity, the TTP223P tactile sensor for counting bee entries and exits, and a load cell for monitoring hive weight. These sensors are directly connected to the NodeMCU, which processes the raw data and prepares it for transmission. The processed data is then sent to the cloud infrastructure using the Blynk platform, which provides a reliable and scalable solution for remote data storage and management. The Blynk cloud enables real-time data streaming to the Unity application, which acts as the user interface for beekeepers. The Unity application, through API calls to the Blynk platform, retrieves and visualizes the sensor data in an intuitive and interactive dashboard. This comprehensive architecture not only ensures continuous monitoring and timely alerts but also facilitates seamless user interaction, empowering beekeepers to make informed decisions to maintain optimal hive conditions and improve overall bee health and productivity.

IV. RESULTS AND ANALYSIS

The following section discusses results obtained during testing phases across various conditions. We have tested,

- Humidity sensor
- Temperature sensor
- Loadcell sensor
- Tactile sensor
- Buzzer actuator

Out of these, we present the first three sensor test steps and the results. The results for Buzzer could not be depicted due to the output being sound, and the results of tactile sensor have already been tested in extreme conditions before.

1. Humidity time series in extreme conditions

We varied the humidity of the beehive system rapidly in the following way:

- We started with a normal humidity value of 50% by producing appropriate vapor. Using this, we were able to test if the sensor was working in a good condition or not. The result was positive.
- We then let the humidity drop to minimum, which happened to be around 26% at the time. Using this we are able to perform the rapid change in humidity task that is moving from min humidity to max humidity.
- Thus, after it, we rapidly incremented the humidity to the maximum of approx. 94% as fast as possible to check whether the sensor is able to keep up with the change and show the results. The sensor was able to capture this spike in humidity perfectly.
- Then, we maintained this max humidity at max value for some time to check the durability of the sensor and precision over time. The sensor seemed to perform well with both, a high accuracy and precision.
- We then dropped the humidity naturally over time till it reaches the minimum which happened to be 20% at that time. This was done to re-test the sensor if there was any effect of this rapid testing on the normal functioning of the sensor. There was no such visible effect observed.
- Finally, we incremented the humidity back to a natural 40% in a beehive by introducing some water vapor. To make a final check of the sensor performance and the results were satisfactory.

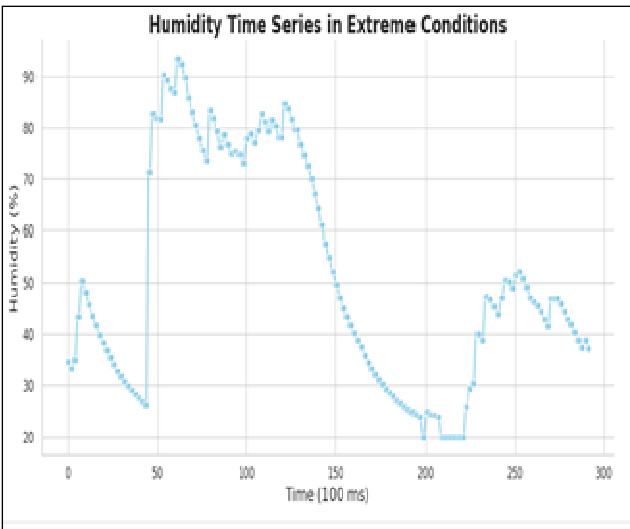


Fig. 6 Graph depicting the changes in Humidity Readings of DHT11 on Extreme Testing

2. Temperature time series in extreme conditions

The temperature sensor was tested in the following way:

- First, we decremented the temperature the minimum possible one, which was about 26 degrees Celsius at the time. This was done to check if there is even a decrease in temperature visible or not. The test showed positive results.

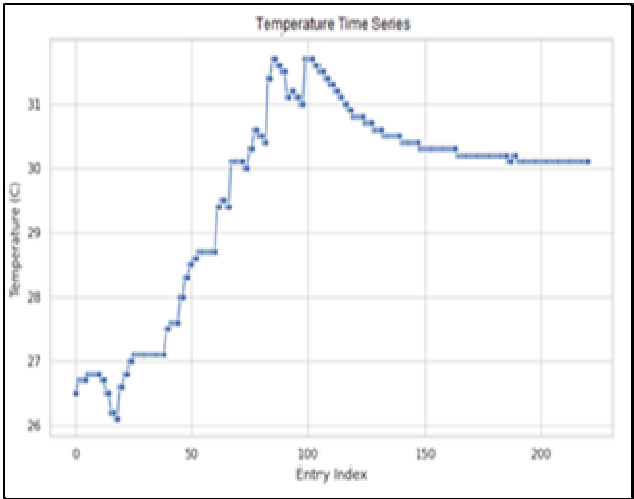


Fig. 7 Graph depicting the changes in Temperature Readings of DHT11 on Extreme Testing

- Next, we rapidly incremented the temperature to the max possible in the system at that time, which was about 32-degree Celsius. We say that the sensor sowed an approximate linear increment in temperature which is a very good indicator that the sensor is accurate and the results of sensor are valid.
- Finally, we let the temperature drop from max to the room temperature naturally. Since this was done naturally without any intervention, the time taken to achieve room temperature was longer than the time needed to reach the max temperature of testing. The temperature sensor captured this slow and steady drop of temperature.

The temperature sensor also reached a constant value at the end which represents the room temperature at that time. This suggests the results of this test were satisfactory.

3. Load/ Weight time series in extreme conditions

We performed the testing of load cell as follows

- We first decremented the mass on loadcell to 0 and calibrated the sensor.
- Then we tested the mass calculation by placing very light weight object on it about 50-100grams. This was done to check if the calibration was correctly done or not.
- Next, the mass on the load cell was incremented extremely from 0 grams to 2 kilo grams+. This was done to check the working of load cells in extreme conditions. The sensor successfully sensed this steep increment and was able to respond to it via great accuracy.

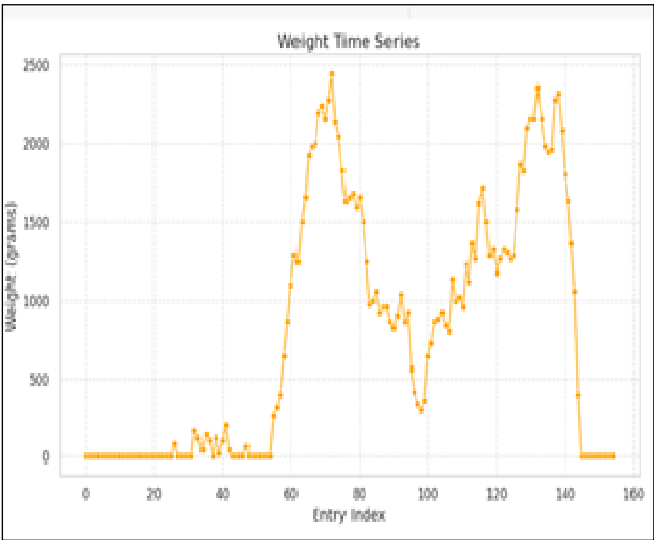


Fig. 8 Graph depicting the changes in the Load Cell Sensor Readings on Extreme Testing

- We then removed the 2+ kilograms of mass slowly part by part, and checked whether the sensor was able to capture the removal of mass or not. The sensor did capture it.
- Then, to make sure the extreme increment in mass didn't cause any issues in the load cell, we incremented the mass on it again to 2+ kilograms and obtained similar results to the first time. This proved the sensor was accurate and precise both.
- Finally, we removed all the mass at once to see how the load cell would react and it did react by making the mass readings as zero with a drop.

❖ Overall System Analysis

The analysis of the beehive weight graph revealed a maximum hive weight of approximately 2.5 kg. This data signifies significant fluctuations in hive weight, indicating potential variations in honey production or hive activity over time. Additionally, the graph displays periodic spikes and dips, suggesting periods of increased foraging activity or colony growth followed by periods of reduced activity or resource consumption. These insights gleaned from the hive weight graph provide valuable information for beekeepers to assess hive health, productivity, and overall colony dynamics.

❖ Application and Importance of Authentication

The application screenshots included demonstrate the login page and a graph page showcasing hive humidity data. Authentication plays a crucial role in ensuring the security and integrity of the application. By requiring users to authenticate themselves before accessing sensitive data and functionalities, such as hive monitoring and management, the app safeguards against unauthorized access and potential misuse. This authentication

mechanism ensures that only authorized personnel, such as beekeepers or researchers, can access and interact with the app's features, thereby enhancing data privacy and overall security.



Fig. 9 Authentication Mechanism for Beehive Monitoring System



Fig. 10 Beehive Monitoring System Application User Interface for Remote Monitoring of Humidity through API calls

❖ Buzzer Ringing Feature for Alert

One notable feature of the application that we added is the buzzer ringing functionality for alerts. This feature enables beekeepers to receive immediate notifications in case of critical events or anomalies detected within the hive, such as sudden weight fluctuations or unusual activity patterns. The ability to toggle the buzzer on or off at will provides users with flexibility in managing alert notifications based on their preferences and specific monitoring requirements. By leveraging this feature, beekeepers can promptly respond to potential threats or emergencies, ensuring timely intervention and mitigation of risks to hive health and productivity.

V. FUTURE SCOPE

Looking ahead, the beehive monitoring app has a lot of room to grow and improve. For starters, we can make it work hand in hand with bigger systems that manage entire farms. Imagine if beekeeping data could be combined with info about crops and weather - that'd give farmers a complete picture of what's happening on their land.

There are quite a many advantages to integrating the Smart Beehive Monitoring System with farm management systems. Beekeepers can correlate bee activity with crop health, optimize hive management based on weather conditions, inventory control, and identify early signs of pest infestation or disease by integrating with crop monitoring and pest/disease control systems. Extensive analysis of environmental elements and bee population trends is made possible by integration with data analytics platforms. Bee colony health and operational efficiency are improved through the automation of hive management duties through remote monitoring and control integration.

We can also make the app smarter by connecting it to more kinds of sensors. This means it could keep an eye on things like air quality and soil moisture, not just bees. Plus, we can teach it to spot patterns in all the data it collects, using things like machine learning. That way, it could predict problems before they happen, like if a hive is getting sick.

Another goal is to make the app more affordable and easier to use. By simplifying the tech it needs and finding cheaper ways to store data, we can make sure more people can benefit from it. And let's not forget about making it user-friendly - with clear screens and options that suit each beekeeper's needs. These improvements could turn the beehive monitoring app into an essential tool for farmers everywhere, helping them keep their bees healthy and their crops thriving.

VI. CONCLUSION

We conclude this research work by summarizing the overall system and highlighting some of its novel features. Our proposed Beehive Monitoring System is capable of remote monitoring of beehives using the IoT technology.

We make use of API calling and Cloud Computing for the connection between the Application and the deployed hardware system. The system is cost effective and allows sensing of four different environmental factors which are crucial for bees. The proposed system alerts users in case of anomalies which includes, abnormal behavior of the bees due to changes in temperate or humidity. It also alerts users when the beehive weight reaches an optimal value for harvesting. Thus, using this system and the associated application, users can enjoy improvement in the overall harvest and control over the beehives. This includes quick decision making ability in cases of anomaly.

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